

Htet Myat Aung (Alexander E. Hong)

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Education

National Chung Cheng University (CCU) , Mechanical Engineering (Automation & Control) - Fully Funded Scholarship - Department: Mechanical Engineering - Program begins 1 September 2026; expected graduation 2028	Chiayi, Taiwan Sept 2026 – 2028
King Mongkut's University of Technology Thonburi (KMUTT) , Automation Engineering - Graduated September 2025 - Thesis: Natural Gas Methane Leak Detection Using an Infrared Camera - Department: Control System and Instrumentation (INC) - INC Department Outstanding Student Scholarship - Winter Semester 2023 & 2024	Bangkok, Thailand 2022 – 2025
Yangon Technological University (Center of Excellence) , Mechatronics Engineering - Participated in Myanmar RoboLeague 2019 as Mentor - Led First Global Challenge 2019 team representing Myanmar	Yangon, Myanmar 2015 – 2022

Experience

Maintenance Technology Center (MTC) - Non-Destructive Detection Technology Lab , Machine Learning Researcher (Full-Time) Developed advanced machine learning solutions for non-destructive testing applications under Prof. Cherdpong Jomdecha, focusing on acoustic emission sensor data analysis and anomaly detection. - Journal Publication (co-author): Contributed MIECT probe development and signal processing for high-speed railway defect inspection - detecting cracks as small as 0.5 mm at speeds up to 30 km/h on ferromagnetic rail steel (published in Nondestructive Testing and Evaluation) - Production Pipeline Engineering: Architected and maintained scalable ETL pipelines in Python to process high-volume acoustic emission sensor data - Workflow Automation: Optimized end-to-end ML workflows, reducing data-processing latency and accelerating model-training lifecycles through automated scripting and rigorous version control - Model Reliability & Scalability: Engineered robust anomaly-detection systems for industrial pipe inspection, focusing on reproducibility and scalable deployment in Linux-based environments - Data Quality Assurance: Implemented automated validation within the data-ingestion pipeline to ensure high-fidelity input for unsupervised clustering models	Bangkok, Thailand Sept 2025 – May 2026 9 months
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Maintenance Technology Center (MTC) - Non-Destructive Detection Technology Lab,

Machine Learning Intern

Independently designed and implemented unsupervised machine learning solutions for GRP pipe inspection using acoustic emission sensor data, demonstrating strong problem-solving and analytical skills in identifying patterns from complex, real-world infrastructure data on Linux based WSL2 platform with version control via Git.

- Won First Runner-Up Award in 'Smart Infrastructure for Smart City' competition by Metropolitan Waterworks Authority
- Listed in KMUTT Hall of Fame for outstanding achievement in Smart Infrastructure development
- Applied AI/ML techniques to solve real-world infrastructure challenges

Bangkok, Thailand

June 2025 – Aug 2025

3 months

HoneyHunt.AI, Junior Machine Learning Intern / Large Language Model Intern

Enhanced automated resume parsing and intelligent job matching systems by integrating AI APIs (advanced LLM APIs), improving accuracy and scalability.

- Improved algorithm scalability and matching accuracy for production-level deployment
- Contributed to product development in AI-powered recruitment technology
- Explored Generative AI techniques to design agentic workflows for automated data processing and decision-making, leveraging LLMs and prompt engineering to optimize AI-driven recruitment solutions
- Implemented API management best practices including versioning, rate limiting, and authentication using API gateways to support reliable deployment of GenAI workflows in production environments

Bangkok, Thailand

Dec 2024 – Feb 2025

3 months

Research

Department of Control Systems and Instrumentation (INC), KMUTT, Computer Vision (Student) Researcher

Developed an optimized 3D-CNN for real-time infrared video classification achieving 99.75% accuracy in methane gas-leak detection from temporal-spatial thermal-camera features.

- Built a classical image-processing pipeline (Gaussian Mixture Models, Running Average, adaptive background subtraction) for robust foreground segmentation under varying lighting conditions
- Optimized the model for edge deployment via input-resolution scaling and quantization, maintaining 99%+ accuracy while reducing inference time by 60%
- First-authored the resulting study, awarded Best Paper at IEEE ICOT 2025

Bangkok, Thailand

June 2024 – Sept 2025

CODIA Lab, KMUTT & Panasonic Singapore, Perception (Student) Research Engineer

Developed and optimized SLAM Sparse Bundle Adjustment algorithms in C for visual odometry applications for real-time edge deployment on ARM-based embedded platforms.

- Optimized SLAM Sparse Bundle Adjustment algorithms for edge devices, improving computational efficiency and enabling real-time performance in resource-constrained environments

Bangkok, Thailand

Feb 2024 – July 2024

COSIN Lab, INC Department, KMUTT, FormulaStudent Vehicle Control Student Researcher

Developed ROS2-based autonomous navigation systems integrating IMU and camera sensor fusion for competitive robotics applications.

- Implemented the micro-ROS framework on microcontrollers for real-time IMU data acquisition and sensor synchronization, achieving sub-10ms latency for control loop feedback
- Performed sensor calibration including IMU bias estimation and camera extrinsic alignment to improve localization accuracy in dynamic environments

Bangkok, Thailand

Nov 2023 – Feb 2024

Volunteer

FIRST GLOBAL CHALLENGE 2019, Mentor

Provided technical guidance and strategic mentorship to Myanmar national robotics team, fostering knowledge sharing through structured training sessions and collaborative problem-solving.

- Led team to win Judges Award at international robotics competition
- Fostered STEM education and international collaboration in robotics

Dubai, UAE

June 2019 – Oct 2019

Laboratory of Robotics and Artificial Intelligence (L.O.R.A), Robotics Researcher and Founder

Founded and led comprehensive robotics research laboratory focusing on practical applications.

- Spearheaded development of autonomous disinfection robots during COVID-19 pandemic in Myanmar
- Managed multiple concurrent research initiatives and mentored junior researchers
- Managed cross-functional communication between engineering teams, public health stakeholders, and external partners to align technical development with societal needs during the pandemic

Yangon, Myanmar

Dec 2019 – June 2021

ConceptX International School, Head of Department and Instructor of Robotics and Computer Science

Instructed Computer Science and Robotics to approximately 200 students from Upper Primary 5 to Lower Secondary 2.

- Promoted to Head of Department in October 2021
- Taught IGCSE Computer Science to Secondary 3 students

Yangon, Myanmar

June 2021 – Dec 2021

Awards

Best Paper Award - IcoICT 2025 (IEEE Xplore)

Co-winner with Asst. Prof. Dr. Sarawan Wongsu

13th International Conference on Information and Communication Technology, Bandung

July 2025

Best Presentation Award - ICA-SP Con 2025

16th Instrumentation, Control and Automation Senior Project Conference

Apr 2025

First Runner-Up - Smart Infrastructure for Smart City Competition

Metropolitan Waterworks Authority, Bangkok

July 2025

KMUTT Hall of Fame

King Mongkut's University of Technology Thonburi

July 2025

INC Department Outstanding Student Scholarship

KMUTT - Control System and Instrumentation Department

2024

INC Department Outstanding Student Scholarship

KMUTT - Control System and Instrumentation Department

2023

Judges Award - First Global Challenge 2019

International Robotics Competition, Dubai, UAE

first.global/2019-nations/myanmar-2019

Oct 2019

Publications

Development of high-speed inspection technique based on motion-induced eddy current testing for surface railway defects

Co-author - contributed MIECT probe development and signal processing. Q1-ranked journal (Web of Science, JCR) publication on a high-speed motion-induced eddy current testing technique for detecting surface railway defects, developed with the Maintenance Technology Center (MTC), KMUTT. DOI - 10.1080/10589759.2026.2638558

N. Brienza, N. Khunsombutcharoen, B. Poopat, C. Janya-anurak, C. Jirarunsatian, R. Mutthanu, Htet Myat Aung, C. Jomdecha

www.tandfonline.com/doi/abs/10.1080/10589759.2026.2638558

Optimized 3D-CNN for Real-Time Infrared Natural Gas Leak Classification

First author - developed a high-accuracy (99.75%) 3D CNN model for methane gas leak detection using infrared imaging. Winner of the Best Paper Award. DOI - 10.1109/ICoICT66265.2025.11192973

Htet Myat Aung, Sarawan Wongsu

ieeexplore.ieee.org/document/11192973

Natural Gas Methane Leak Detection Using an Infrared Camera

Best Presentation Award winner for research on natural gas leak detection systems using infrared imaging technology.

Htet Myat Aung

Skills

Perception & Vision

Robotics

Programming Languages

ML/DL

Tools

Languages

English

Fluent (TOEFL 84 - Writing: 25, Speaking: 26 & Extensive technical documentation and international collaboration experience)

Burmese

Native

Interests

Artificial Intelligence & Machine Learning

Robotics & Automation

Research & Innovation